

Derivative of a function



Outline

Line through two points

Tangent line

Derivative of a function at a point and derivative function

Continuity and differentiability

Line through two points

Introduction: Line through two points

- ▶ In this section we derive the equation of the **line through two points** of the plane.
- ▶ We introduce the **slope** m and the **y -intercept** q of a line.
- ▶ The slope of a line through two points is the starting point to arrive at the **tangent line** and hence at the **derivative**.

Equation of the line in explicit form

- ▶ Starting from two points

$$(x_1, y_1) \text{ and } (x_2, y_2)$$

it is possible to compute the function

$$y = m x + q$$

whose graph is the line through the two points (equation of the line in explicit form).

- ▶ The **slope** of the line is the value m and the **y-intercept** of the line is the value q .

Computing m and q (1/2)

- Requiring the line to pass through the two points we have:

$$\begin{cases} y_1 = m x_1 + q \\ y_2 = m x_2 + q \end{cases}$$

and we can determine m and q as follows (two equations in two unknowns):

$$\begin{cases} q = y_1 - m x_1 \\ q = y_2 - m x_2 \end{cases} \quad \begin{cases} q = y_1 - m x_1 \\ y_1 - m x_1 = y_2 - m x_2 \end{cases}$$

where the last equation becomes $m x_2 - m x_1 = y_2 - y_1$.

Computing m and q (2/2)

$$\begin{cases} q = y_1 - m x_1 \\ m = \frac{y_2 - y_1}{x_2 - x_1} \end{cases} \quad \begin{cases} q = y_1 - \frac{y_2 - y_1}{x_2 - x_1} x_1 \\ m = \frac{y_2 - y_1}{x_2 - x_1} \end{cases}$$

And we obtain the line:

$$y = \underbrace{\frac{y_2 - y_1}{x_2 - x_1}}_m x + \underbrace{y_1 - \frac{y_2 - y_1}{x_2 - x_1} x_1}_q$$

clearly

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2}$$

Line through a point with given slope

The line through two points (x_1, y_1) and (x_2, y_2) :

$$y = \underbrace{\frac{y_2 - y_1}{x_2 - x_1}}_m x + y_1 - \underbrace{\frac{y_2 - y_1}{x_2 - x_1}}_m x_1$$

can be rewritten as $y = m x + y_1 - m x_1$.

Hence the line can be written only in terms of the slope m and the point (x_1, y_1) as follows:

$$y = y_1 + m(x - x_1) \quad \text{with} \quad m = \frac{y_1 - y_2}{x_1 - x_2}$$

The same holds for any point $(\tilde{x}, \tilde{y})$ of the line; in this case we would have:

$$y = \tilde{y} + m(x - \tilde{x})$$

Example: line through two points (1/2)

We compute the line through the points: $(x_1, y_1) = (-1, -1)$ and $(x_2, y_2) = (\frac{1}{2}, \frac{1}{8})$.

We compute the slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\frac{1}{8} + 1}{\frac{1}{2} + 1} = \frac{3}{4}$$

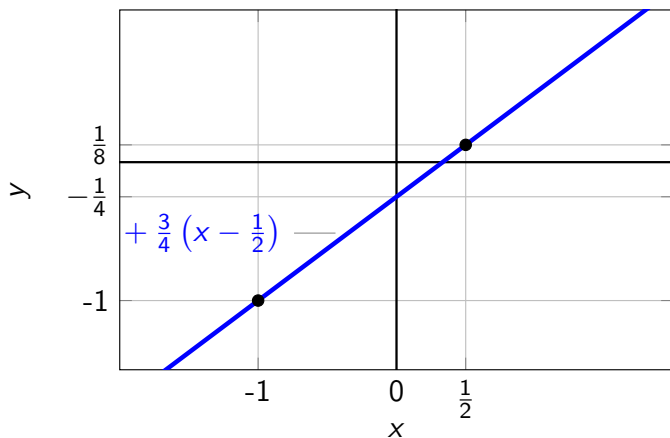
Using the point $(x_2, y_2) = (\frac{1}{2}, \frac{1}{8})$ we obtain:

$$y = \frac{1}{8} + \frac{3}{4} \left(x - \frac{1}{2} \right) = \frac{3}{4} x - \frac{1}{4}$$

Now using the point $(x_1, y_1) = (-1, -1)$ we obtain the same line:

$$y = -1 + \frac{3}{4} (x + 1) = \frac{3}{4} x - \frac{1}{4}$$

Example: line through two points (2/2)



Summary

Line through two points:

- ▶ Equation in explicit form: $y = mx + q$, with m the slope and q the y -intercept
- ▶ Slope of the line through (x_1, y_1) and (x_2, y_2) : $m = \frac{y_2 - y_1}{x_2 - x_1}$
- ▶ Line through a point $(\tilde{x}, \tilde{y})$ with slope m : $y = \tilde{y} + m(x - \tilde{x})$

Tangent line

Introduction: Tangent line

- ▶ In this section we ask what the **tangent line** to the graph of a function at a point is.
- ▶ We will see that the intuitive definition (“touches without cutting”) does not always work.
- ▶ We will arrive at the idea of the tangent as the **limit position** of lines through two points of the graph, and we will see why the tangent is useful in practice (points with **horizontal tangent**).



What is the tangent line?

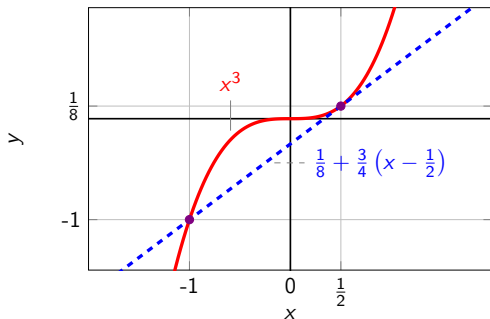
What is the **tangent** line to the graph of a function at a point?

- ▶ The intuitive definition of a line tangent to the graph is that of a line that “touches” the graph without “cutting” or “crossing” it (imagining the curve as if it were an impenetrable physical object).
- ▶ This intuitive definition works, for example, for parabolas or hyperbolas.

The intuitive definition is not enough (1/2)

However, take for example the function $f(x) = x^3$:

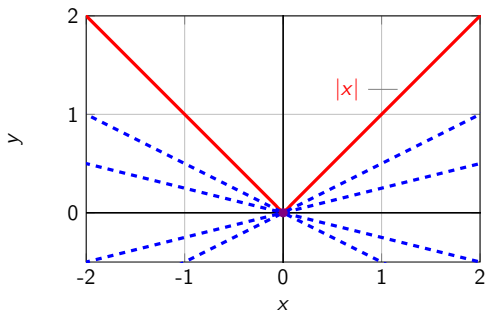
the graph $y = x^3$ and the line $y = \frac{1}{8} + \frac{3}{4} \left(x - \frac{1}{2} \right)$



The intuitive definition works at the point $(\frac{1}{2}, \frac{1}{8})$, but the line also intersects the graph of the function at the point $(-1, -1)$.

The intuitive definition is not enough (2/2)

Conversely, there are functions, such as $f(x) = |x|$, whose graph is $y = |x|$:



This graph has a point (the origin) at which there are infinitely many lines satisfying the intuitive definition, but clearly none of them can be called tangent.

The tangent as a limit line

Geometrically, the tangent line can be approximated by the **line through two points** very close to each other on the curve, the first one **fixed** (the point of tangency) and the other one “**moving**” (and chosen closer and closer to the point of tangency).

- ▶ Two points determine a line; the closer the two points get, the closer this line gets to the tangent.
- ▶ We therefore need to understand what happens to the line through the two points when the second, moving point gets closer and closer to the first one, without ever coinciding with it. The “**limit line**” – if it exists – will be called the **tangent line**.

The tangent as the best approximation

A further way of looking at the concept of tangency is to think of the tangent (to a curve at a point) as the line that **best approximates** the curve in a neighborhood of the point.

Practical use of the tangent line

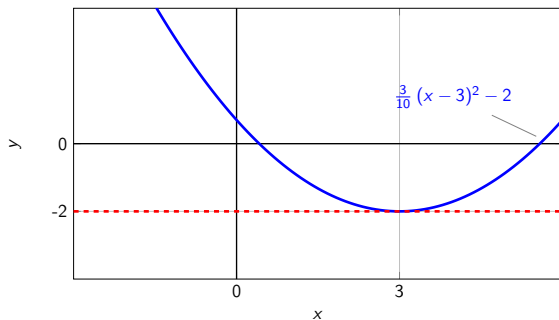
Computing the tangent is important to determine the points where the graph of a function has a **horizontal tangent** (local or global maximum and minimum points, and possibly other points as well).

- ▶ It is therefore useful to be able to write analytically the equation of the tangent line to the curve at a generic point, and then see at which points it is horizontal.
- ▶ This idea is due first to Fermat, who developed it around 1630.

Example: point with horizontal tangent

Consider the function:

$$f(x) = \frac{3}{10} (x - 3)^2 - 2$$

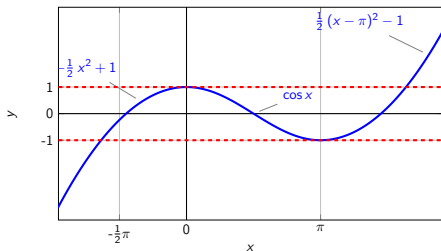


The function has a point with horizontal tangent, which is a global minimum point.

Example: points with horizontal tangent (1/2)

Consider the (piecewise-defined) function:

$$f(x) = \begin{cases} -\frac{1}{2}x^2 + 1 & x \leq 0 \\ \cos x & 0 < x \leq \pi \\ \frac{1}{2}(x - \pi)^2 - 1 & x > \pi \end{cases}$$

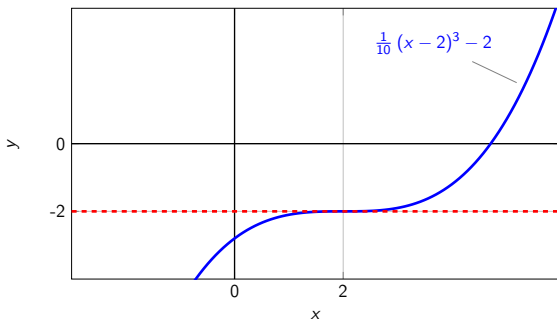


The function has two points with horizontal tangent, which are local minimum and maximum points, for example in the interval $[-\frac{1}{2}\pi, \pi]$.

Example: points with horizontal tangent (2/2)

Now consider the function:

$$f(x) = \frac{1}{10} (x - 2)^3 - 2$$



The function has a point with horizontal tangent which, however, is neither a maximum nor a minimum point.

Summary

Tangent line:

- ▶ The intuitive definition (“touches without cutting”) does not always work: see x^3 and $|x|$
- ▶ The tangent is the **limit line** of the lines through two points of the graph, one fixed and one moving towards the first
- ▶ The tangent is the line that **best approximates** the curve near the point
- ▶ Points with **horizontal tangent** can be minimum points, maximum points, or neither

Derivative of a function at a point and derivative function



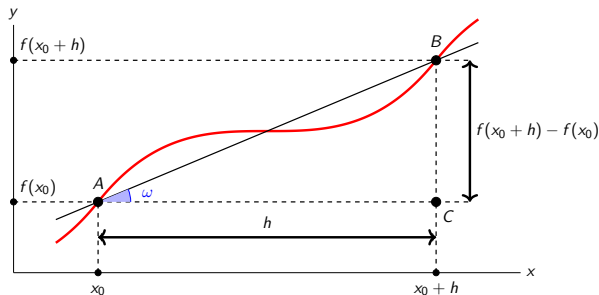
Introduction: Derivative at a point and derivative function

- ▶ **Differential calculus** is the study of the notion of derivative, and it is part of infinitesimal calculus.
- ▶ In this section we look for the **equation of the tangent line** to the graph of the function at a **generic point** A of its graph.
- ▶ We will thus arrive at the **definition of derivative** as the limit of the **difference quotient**, at the tangent line and at the **derivative function**.

Derivative and tangent line

- Consider the graph of a generic function $y = f(x)$; the coordinates of the points A and B in the figure are respectively:

$$(x_0, f(x_0)) \quad \text{and} \quad (x_0 + h, f(x_0 + h))$$



Difference quotient

- ▶ The **slope** of the line through the point A and the point B is:

$$\frac{f(x_0 + h) - f(x_0)}{h}$$

Note that a positive slope indicates (on average) a rise, while a negative one indicates a fall (on average).

This ratio is called the **difference quotient** of the function f relative to the interval $[x_0, x_0 + h]$.

- ▶ Geometrically, since the triangle ABC is a right triangle, we have:

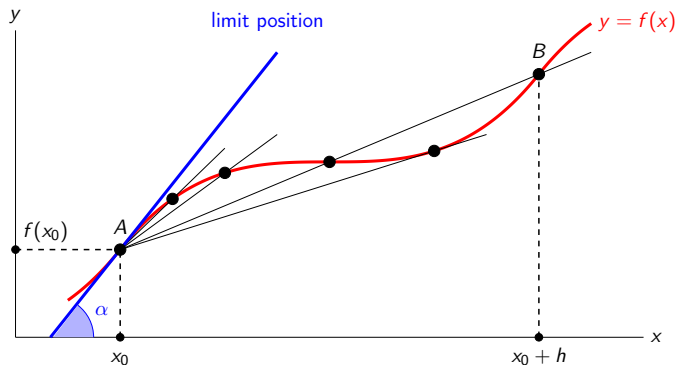
$$\frac{f(x_0 + h) - f(x_0)}{h} = \tan \omega$$

(**slope** of the line through A and B).

Taking the limit

- ▶ Consider the difference quotient and take the limit (assuming it exists) as $h \rightarrow 0$. Geometrically we have:
 1. the point A , with coordinates $(x_0, f(x_0))$, stays fixed
 2. while the point B , with coordinates $(x_0 + h, f(x_0 + h))$, moves towards A (staying on the graph of f).
- ▶ Moving B towards A along the graph of the function, the line AB changes its slope and settles into a **limit position**.

Limit position



Limit line and first derivative

1. the **limit line** is called the **tangent line to the graph** of the function f at the point with abscissa x_0 ;
2. its **slope** is given by $\tan \alpha$ and is called the **first derivative** of the function f at the point x_0 .
3. The angle α is the angle between the tangent line and the x -axis.

Definition 1: of derivative

Let $f : (a, b) \rightarrow \mathbb{R}$; f is said to be differentiable at $x_0 \in (a, b)$ if the following limit exists and is finite

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

this limit is called the first derivative (or simply the derivative) of f at x_0 .

With $h = x - x_0$, we have $h \rightarrow 0 \Leftrightarrow x - x_0 \rightarrow 0 \Leftrightarrow x \rightarrow x_0$ and $x = x_0 + h$, hence:

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Notations for the derivative

- To denote the derivative the following symbols are used:

$\underbrace{f'(x_0)}$
Lagrange's notation

$\underbrace{\dot{f}(x_0)}$
Newton's notation

$\underbrace{\left. \frac{df}{dx} \right|_{x=x_0} \quad \text{and} \quad \left. \frac{dy}{dx} \right|_{x=x_0}}_{\text{Leibniz's notation}}$

Tangent line

Definition 2: tangent line

Given a function $f : (a, b) \rightarrow \mathbb{R}$ with f differentiable at $x_0 \in (a, b)$, the line with equation:

$$y = f(x_0) + f'(x_0)(x - x_0)$$

is called the **tangent line** to the graph of the function f at the point $(x_0, f(x_0))$.

Example: tangent line (1/2)

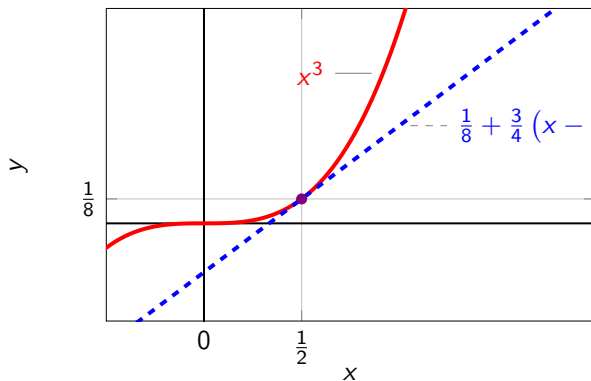
We compute the equation of the tangent line to the graph of the function $f(x) = x^3$ at the point with abscissa $x_0 = \frac{1}{2}$. The derivative at the point $x_0 = \frac{1}{2}$ is:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{\left(\frac{1}{2} + h\right)^3 - \left(\frac{1}{2}\right)^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{8} + \frac{3}{4}h + \frac{3}{2}h^2 + h^3 - \frac{1}{8}}{h} \\ &= \lim_{h \rightarrow 0} \left(\frac{3}{4} + \frac{3}{2}h + h^2 \right) = \frac{3}{4}\end{aligned}$$

We have $f\left(\frac{1}{2}\right) = \frac{1}{8}$, $f'\left(\frac{1}{2}\right) = \frac{3}{4}$ and the tangent line at the point $\left(\frac{1}{2}, \frac{1}{8}\right)$ is:

$$y = f(x_0) + f'(x_0)(x - x_0) = \frac{1}{8} + \frac{3}{4}\left(x - \frac{1}{2}\right)$$

Example: tangent line (2/2)



Example: tangent line (1/2)

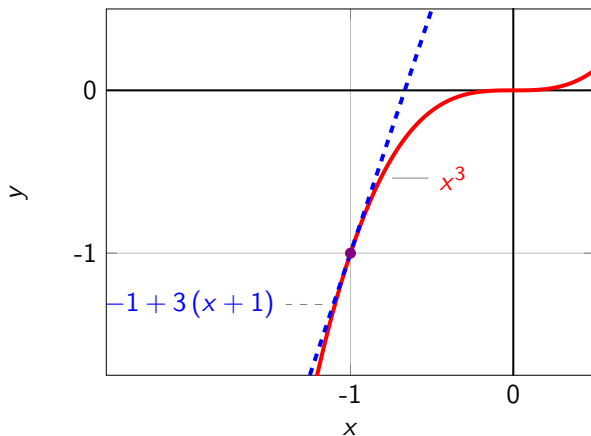
We compute the equation of the tangent line to the graph of the function $f(x) = x^3$ at the point with abscissa $x_0 = -1$. The derivative at the point $x_0 = -1$ is:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} &= \lim_{h \rightarrow 0} \frac{(-1 + h)^3 - (-1)^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1 + 3h - 3h^2 + h^3 + 1}{h} \\ &= \lim_{h \rightarrow 0} (3 - 3h + h^2) = 3\end{aligned}$$

We have $f(-1) = -1$, $f'(-1) = 3$ and the tangent line at the point $(-1, -1)$ is:

$$y = f(x_0) + f'(x_0)(x - x_0) = -1 + 3(x + 1)$$

Example: tangent line (2/2)



Derivative function

- We can now define a function that associates to each x the derivative of a function f at the point x (of course, if f is differentiable).

Definition 3: of derivative function

If a function f is differentiable at every point of an interval (a, b) , the function:

$$f' : (a, b) \rightarrow \mathbb{R}, \quad f' : x \mapsto f'(x)$$

*is called the **derivative function** of f .*

Notations for the derivative function

- To denote the first derivative function the following notations are used:

$f'(x)$
Lagrange's notation

$\dot{f}(x)$
Newton's notation

$\frac{df}{dx}$, $\frac{df(x)}{dx}$ and $\frac{dy}{dx}$
Leibniz's notation

Example: derivative function (1/5)

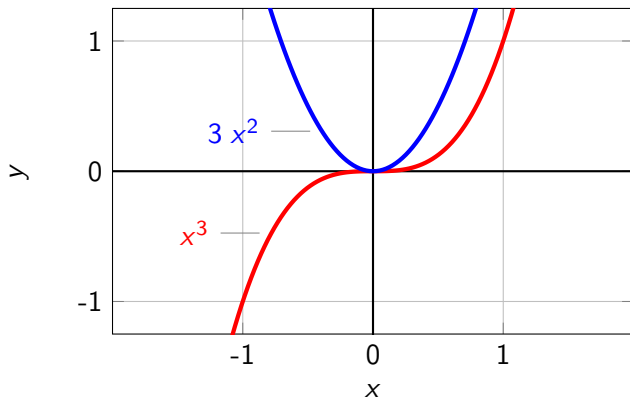
We compute the derivative function of the function $f(x) = x^3$. For every $x \in \mathbb{R}$ we have:

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - (x)^3}{h} \\&= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\&= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2\end{aligned}$$

hence

$$f'(x) = 3x^2$$

Example: derivative function (2/5)



Example: derivative function (3/5)

We compute the derivative function of the function $f(x) = |x|$. For $x \neq 0$, we have:

► if $x > 0$: $f(x) = x$ and

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = 1$$

hence $f'(x) = 1$

► if $x < 0$: $f(x) = -x$ and

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h)+x}{h} = -1$$

hence $f'(x) = -1$

Example: derivative function (4/5)

For $x = 0$ we have:

$$\frac{f(x+h) - f(x)}{h} = \frac{f(h) - f(0)}{h} = \frac{|h|}{h}$$

hence

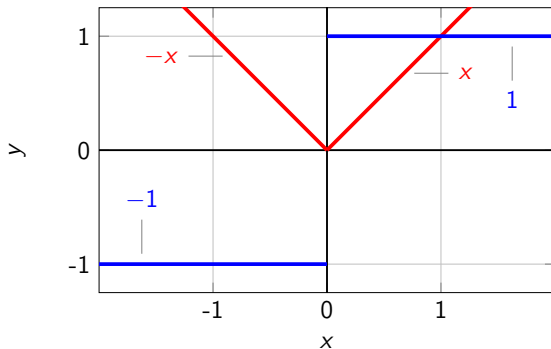
$$\lim_{h \rightarrow 0^+} \frac{|h|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1, \quad \lim_{h \rightarrow 0^-} \frac{|h|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

since if $h \rightarrow 0^+$ then $|h| = h$ and if $h \rightarrow 0^-$ then $|h| = -h$.

We conclude that, since the limit of the difference quotient does not exist, f is not differentiable at $x = 0$.

Example: derivative function (5/5)

The function is continuous at the origin but the tangent is not well defined.



Derivative and derivative function:

- ▶ **Difference quotient:** $\frac{f(x_0+h)-f(x_0)}{h}$, slope of the line through A and B
- ▶ **Derivative** at x_0 : finite limit of the difference quotient as $h \rightarrow 0$
- ▶ **Tangent line** at $(x_0, f(x_0))$: $y = f(x_0) + f'(x_0)(x - x_0)$
- ▶ **Derivative function:** $f' : x \mapsto f'(x)$; Lagrange's, Newton's and Leibniz's notations
- ▶ $f(x) = x^3$ has derivative $3x^2$; $f(x) = |x|$ is not differentiable at $x = 0$

Continuity and differentiability

Introduction: Continuity and differentiability

- ▶ In this section we study the relation between **differentiability** and **continuity** at a point.
- ▶ We will prove that a function differentiable at a point is also continuous at that point.
- ▶ We will see that the converse is false (counterexample: $|x|$), and how to use the result to rule out differentiability at points of discontinuity.

Differentiability implies continuity

Theorem 1

Given a function $f : [a, b] \rightarrow \mathbb{R}$, if f is differentiable at a point $x_0 \in (a, b)$ then it is continuous at x_0 .

Proof

The function is differentiable at $x_0 \in (a, b)$, so we write:

$$f(x_0 + h) - f(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} \cdot h \sim f'(x_0) \cdot h \quad \text{for } h \rightarrow 0$$

Moreover we have

$$f'(x_0) \cdot h \rightarrow 0 \quad \text{for } h \rightarrow 0$$

Therefore

$$\lim_{h \rightarrow 0} (f(x_0 + h) - f(x_0)) = 0 \quad \text{hence} \quad \lim_{h \rightarrow 0} f(x_0 + h) = f(x_0)$$

Setting $x_0 + h = x$ we have $h \rightarrow 0$ if and only if $x \rightarrow x_0$, and consequently we have:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$



Alternative proof

A possible alternative but equivalent proof is the following.

The function is differentiable at $x_0 \in (a, b)$, so we write:

$$\begin{aligned}\lim_{h \rightarrow 0} f(x_0 + h) &= \lim_{h \rightarrow 0} f(x_0 + h) - f(x_0) + f(x_0) \\ &= \lim_{h \rightarrow 0} \underbrace{\frac{f(x_0 + h) - f(x_0)}{h}}_{\rightarrow f'(x_0)} \cdot \underbrace{h}_{\rightarrow 0} + f(x_0) = f(x_0)\end{aligned}$$

Setting $x_0 + h = x$ we have $h \rightarrow 0$ if and only if $x \rightarrow x_0$, and consequently we have:

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$



The converse is not true

Given a function f , differentiability at a point x_0 in the interior of its domain implies continuity at the point x_0 :

$$f \text{ is differentiable at } x_0 \Rightarrow f \text{ is continuous at } x_0$$

but the converse is not true:

$$f \text{ is continuous at } x_0 \not\Rightarrow f \text{ is differentiable at } x_0$$

a counterexample is the function $f(x) = |x|$, which is continuous at $x_0 = 0$ but not differentiable at $x_0 = 0$.

Consequently, if a function is continuous at x_0 , it is not necessarily differentiable at x_0 .

Necessary and sufficient conditions

- ▶ The fact that f is differentiable at x_0 is a sufficient but not necessary condition for f to be continuous at x_0 .
- ▶ Moreover, the fact that f is continuous at x_0 is a necessary but not sufficient condition for f to be differentiable at x_0 .
- ▶ From the contrapositive of the implication “differentiable $\Rightarrow$ continuous” we have:

f is not continuous at $x_0 \Rightarrow f$ is not differentiable at x_0

that is, if a function is discontinuous at x_0 it cannot be differentiable at x_0 .

Summary

Continuity and differentiability:

- ▶ If f is differentiable at x_0 then f is continuous at x_0
- ▶ The converse is false: $|x|$ is continuous but not differentiable at 0
- ▶ If f is discontinuous at x_0 then it is not differentiable at x_0

Key messages

Summary

1. From the secant to the tangent:

- ▶ The slope of the line through two points of the graph is the **difference quotient**
- ▶ The **tangent line** is the limit position of the secant lines

2. Derivative:

- ▶ $f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$ (if it exists and is finite)
- ▶ Tangent line: $y = f(x_0) + f'(x_0)(x - x_0)$

3. Continuity:

- ▶ Differentiable $\Rightarrow$ continuous, but not conversely